



GATE SMASHERS

**GATE2024**

**Engineering Mathematics**

**Topic: Probability and Statistics**

Subtopic: .....Counting (Basic Permutation).....

## LECTURE - I

EXPLANATION

**Fundamental Principle Of Counting -** If an event A can occur in  $m$  different ways, following which another event B (Independent of A) can occur in  $n$  different ways.



Total occurrence of both events A and B in given order is  $m \times n$

Similarly for three events =  $m \times n \times w$

### Example -

1) There are four routes between Delhi to Goa. In how many different ways can a man go from Delhi to Goa and return, if for returning

- i) Any of the route is taken
- ii) The same route is not taken

2) How many four digit numbers greater than 4000 can be formed with digits 0, 1, 2, 3, 4 if

- (i) Repetition of digit is allowed.
- (ii) Repetition of digit is not allowed.

3) How many odd number can be formed by using the digit 0, 2, 5, 7 when repetition is not allowed.



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**NOTES PREPARED BY: ANKIT ARORA**